Yingshi Huang

CONTACT INFORMATION

California, United States
+1 310 873 7956
yingshi@ucla.edu
https://quantithinker-yingshi.github.io/

EDUCATION

University of California, Los Angeles (UCLA), California, United States
Ph.D. in Education — Sep. 2023 - Present
M.S. in Statistics & Data Science — Sep. 2024 - Present

- Research Interest: Latent variable modeling; Social networks; Casual inference
- Advisor: Prof. Minjeong Jeon

Beijing Normal University, Beijing, China
M.A. in Psychology — Sep. 2020 – Jun. 2023

South China Normal University, Guangzhou, China
B.S. in Psychology — Sep. 2016 – Jun. 2020

PUBLICATIONS

(* correspondent author, + co-first author)

Educational and Psychological Measurement

1. Zhang, L., Chen, P.*, **Huang, Y.**, Jeon, M. (under review). Uncovering residual structure with variational autoencoders.
2. **Huang, Y.**, Wang, S., Pan, Y., Lu, X., & Chen, P.* (under review). A motivation-based cognitive diagnostic model for disengaged responses detection.
3. Yuan, L., **Huang, Y.**, & Chen, P.* (2025). Online calibration for multidimensional CAT with polytomously scored items: A neural network–based approach. *Journal of Educational and Behavioral Statistics*, *0*(0). doi:10.3102/10769986251315531.
4. Chen, P.*, Dai, Y., **Huang, Y.** (2023). Test mode effect: Sources, detection, and applications (in Chinese). *Advances in Psychological Science*, *31*(10), 1966–1980. doi: 10.3724/SP.J.1042.2023.01966
5. Yuan, L., **Huang, Y.**, Li, S., & Chen, P.* (2023). Online calibration in multidimensional computerized adaptive testing with polytomously scored items. *Journal of Educational Measurement*, *60*(3), 476-500. doi: 10.1111/jedm.12353.
6. **Huang, Y.**, Ren, H., & Chen, P.* (2023). Item selection methods with exposure and time control for computerized classification test. *British Journal of Mathematical and Statistical Psychology*, *76*, 52–68. doi: 10.1111/bmsp.12281.
7. Ren, H., **Huang, Y.**, & Chen, P.* (2022). Types, characteristics and application of termination rules in computerized classification testing (in Chinese). *Advances in Psychological Science*, *30*(5), 1168. doi: 10.3724/SP.J.1042.2022.01168.

Substantive Topics

1. Tien, I., Luo, J., **Huang, Y.**, & Jeon, M.* (under review). Visualizing sex differences on the ADOS-2 using a latent space item response model.
2. Tien, I.*, Wolpe, S., **Huang, Y.**, Sozeri, S., Lee, M., & Jeon, M. (2025). How diagnostically accurate are #Autism portrayals? A latent space item response modeling approach. *Journal of Autism and Developmental Disorders*, 1-14. doi:10.1007/s10803-025-07164-5.
3. **Huang, Y.**, Luo, J., Shetty, V., & Jeon, M.* (2025). The power of social talk: A longitudinal network analysis of conversations in fostering interdisciplinary collaboration. *Journal of Clinical and Translational Science*, 1–26. doi:10.1017/cts.2025.10124.

4. Ren, J., Luo, J., **Huang, Y.**, Shetty, V., & Jeon, M.* (2025). Mapping the mHealth nexus: A semantic analysis of mHealth scholars' research propensities following an interdisciplinary training institute. *Applied Sciences*, *15*(11), 6252. doi: 10.3390/app15116252.
5. Ni, Y.*, **Huang, Y.**, Zhao, S., Ma, Y., & Fang, M. (2025). Development and evaluation of the shortened Chinese version of school social behavior scales-2 using item response theory and psychometric network analysis. *Measurement and Evaluation in Counseling and Development*, 1-19. doi: 10.1080/07481756.2025.2499965.
6. Lu, A.*+, Deng, R.+, **Huang, Y.+**, Song, T., Shen, Y., Fan, Z., & Zhang, J. (2022). The roles of mobile app perceived usefulness and perceived ease of use in app-based Chinese and English learning flow and satisfaction. *Education and Information Technologies*, 1-22. doi: 10.1007/s10639-022-11036-1.
7. Ni, Y., Tein, J. Y., Zhang, M.*, Zhen, F., Huang, F., **Huang, Y.**, Yao, Y., & Mei, J. (2020). The need to belong: A parallel process latent growth curve model of late life negative affect and cognitive function. *Archives of Gerontology and Geriatrics*, *89*, 104049. doi: 10.1016/j.archger.2020.104049.

Conference Presentations

1. **Huang, Y.**, & Jeon, M. (2026) *ADAM estimation for the hierarchical joint model of response accuracy and response time*. Talk will be presented at the Annual Meeting of the American Educational Research Association, Los Angeles, CA.
2. **Huang, Y.**, Luo, J., & Jeon, M. (2026) *Dimension selection guideline for latent space item response models*. Talk will be presented at the Annual Meeting of the National Council on Measurement in Education, Los Angeles, CA.
3. **Huang, Y.**, Luo, J., Molenaar, D., & Jeon, M. (2025, April) *Capturing item preknowledge with a response time latent space model framework*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Denver, CO.
4. **Huang, Y.**, Wang, S., Pan, Y., Lu, X., & Chen, P. (2024, April) *A motivation-based cognitive diagnostic model for insufficient responses detection*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Philadelphia, PA.
5. Yuan, L., **Huang, Y.**, & Chen, P. (2023, July). *Online calibration for P-MCAT: A neural network based approach*. Talk presented at the International Meeting of the Psychometric Society, College Park, Maryland.
6. **Huang, Y.**, Zhang, T., & Chen, P. (2022, July). *Exploring the structure of speed in cognitive diagnostic models*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
7. Zhang, T., **Huang, Y.**, & Xin, T. (2022, July). *An analysis of adaptive learning recommendation based on reinforcement learning*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
8. **Huang, Y.**, Ren, H., & Chen, P. (2022, April). *New item selection designs for computerized classification test*. Flash Talk presented at the Annual Meeting of the National Council on Measurement in Education, San Diego, CA, Virtual Meeting.
9. Ren, H., **Huang, Y.**, & Chen, P. (2021, October). Who has the final say in computerized classification test: three types of termination rules. Talk presented at the 23rd National Academic Conference of Psychology, Inner Mongolia.

Funded Grants

2021 Independent Project (Grant No. BJZK-2020A2-20011). **PI**
Item Selection Methods for Computerized Classification Testing 2021 – 2022
Funded by Collaborative Innovation Center of Assessment for Basic Education Quality $1,000

2020 National Natural Science Foundation of China (Grant No. 32071092). **Participant**
Online Calibration in Computerized Adaptive Testing: New Challenges and Solutions 2021 – 2024

Funded by National Natural Science Foundation of China $90,000

2019 Extracurricular Research Project (Grant No. 19XLGA01). **PI**
High Quality of Teacher-Student Relationships Promotes Mathematics Achievement of Junior High School Students: The Multiple Mediation Effect 2019 – 2020
Funded by South China Normal University $500

2018 Special Funds for Seed Breeding Program (Grant No. 18XLGA07). **Co-PI**
Second Language Learning through Mobile Devices: The Effect of Perceived Usefulness, Perceived Ease of Use, and Flow Experience 2018 – 2019
Funded by South China Normal University $500

Research Experience

Jeon lab, Los Angeles, United States
Graduate Student Researcher Dec. 2023 – Present
Prof. Minjeong Jeon's lab centers on developing, applying, and estimating various latent variable models for studying measurement and growth
- developing cultural relevant reading assessment for visually impaired children
- analyzing the NIH-supported annual mHealth Training project with TERGMs
- developing longitudinal latent process model with multiple learning targets

National Assessment Center for Key Technologies, Beijing, China
Research Assistant Sep. 2020 – Jun. 2023
Prof. Ping Chen's lab focuses on building and developing more effective and efficient algorithms for parameter estimation and adaptive testing
- Designed and conducted studies introducing new item selection methods, online calibration designs, and a motivation-based cognitive diagnosis model
- Helped manage the pretesting of items for the National Assessment

National Assessment Guangdong Sub-Center for Item Bank Construction, Guangzhou, China
Research Assistant Sep. 2017 – Jun. 2020
Prof. Minqiang Zhang's lab studies the validity of new Verbal and Mathematics item formats, the score reports system, and test equating in China's College Entrance Examination
- Contributed to the Test Quality Analysis Report (Math) of China's College Entrance Exam
- Assisted with the implementation of the Guangzhou Sunshine education evaluation project and helped write the survey report
- Designed a study on how students' mathematics performance is affected by the quality of the teacher-student relationship, analyzed data from 3,997 students, and led a funded grant

Guangdong Key Laboratory of Mental Health and Cognitive Science, Guangzhou, China
Data Analysis Research Assistant Oct. 2018 – May. 2019
Prof. Aitao Lu's lab studies the mechanism of language acquisition and how human brains execute language processing
- Designed and conducted studies investigating app-based second-language learning, especially for Chinese and English learners
- Collected 786 questionnaires and analyzed the data with latent profile analysis, hierarchical regression, structural equation modeling, and network analysis

Teaching Experience

UCLA, California, United States. *Teaching Assistant*
EDUC 230C: Linear Statistical Models: Analysis of Designed Experiments Mar. 2025 – Jun. 2025
EDUC 230B: Linear Statistical Models: Multiple Regression Analysis Jan. 2025 – Mar. 2025
EDUC 230A: Introduction to Research Design and Statistics Sep. 2024 – Dec. 2024
EDUC 151: Quantitative Research in Education Sep. 2024 – Dec. 2024; Sep. 2025 – Dec. 2025

Beijing Normal University, Beijing, China. *Teaching Assistant*

Adaptive testing and diagnostic adaptive assessment (graduate course) Sep. 2021 – Jan. 2022
Think and act like a psychometrician (undergraduate course) Feb. 2021 – Jun. 2021

Skills

- **Analytical:** Proficient in R, M*plus*, SPSS; Capable of Python
- **Design:** Photoshop, HTML, LaTeX, Unity

Honors and Awards

Graduate Dean's Scholar Award, UCLA 2024, 2025

Gordon & Olga Smith Scholarship, UCLA 2023

Outstanding Graduates of Beijing, Beijing Municipal Education Commission 2023

Outstanding Graduates, Beijing Normal University 2023

National Scholarship, Minister of Education of China 2022

First-class Academic Scholarship, Beijing Normal University 2021

Second-class Comprehensive Scholarship, South China Normal University 2017 - 2018

Administrative and Other Experience

Team Leader & Psychology Teacher, Shenzhen Hongling Middle School 2019
Organized mental health assessment for 696 students, taught 25 courses, and provided psychological counselling to 36 students (Awarded as Excellent Team Leader)

Vice Chairman, The Business Pioneering Camp 2018 – 2019
Planned and organized business competitions, trained new members, and managed the budget (Awarded as Outstanding Student Leader)

Editorial Assistant, Journal of Educational and Behavioral Statistics (JEBS) Jan. 2026 – Present

Reviewer Experience:
Current Psychology (peer-reviewed journal); National Council on Measurement in Education (NCME) Annual Meeting 2025; American Educational Research Association (AERA) Annual Meeting 2024, 2025, 2026

Volunteer Experience:
Devoted to supporting education at the Wanzi Primary School in Meizhou, an undeveloped county in Southeast China (Awarded as Advanced Individual) [Program Web] Aug. 2017